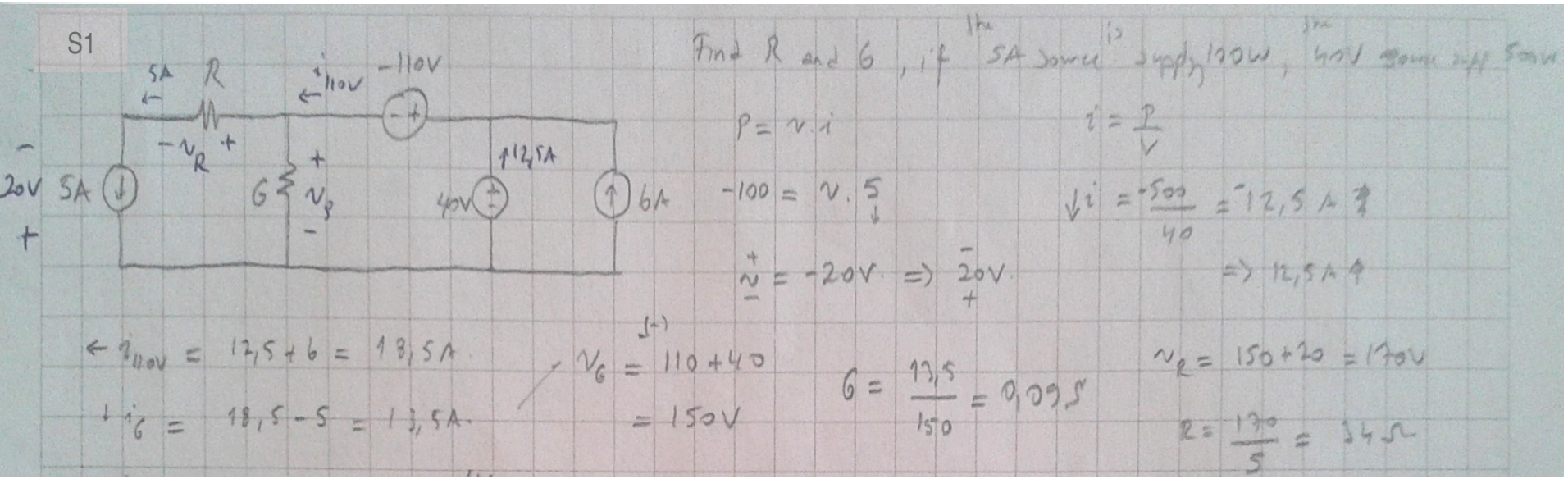
DEVRELER ve SİSTEMLER

BIMU2058 – CSBM2092

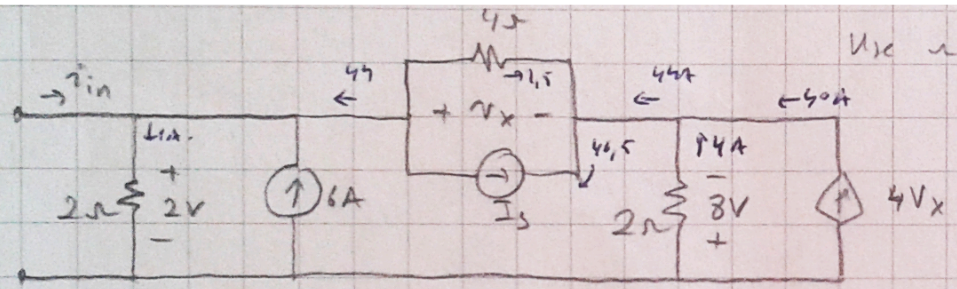
Yrd. Doç. Dr. Fatih KELEŞ

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S2



Use ω and KLs a) v_x b) i_{in} c) I_s

a) $-2 + v_x - 8 = 0$.

$v_x = 2 + 8 = 10V$.

b) $i_{2\Omega} = \frac{2}{2} = 1A$.

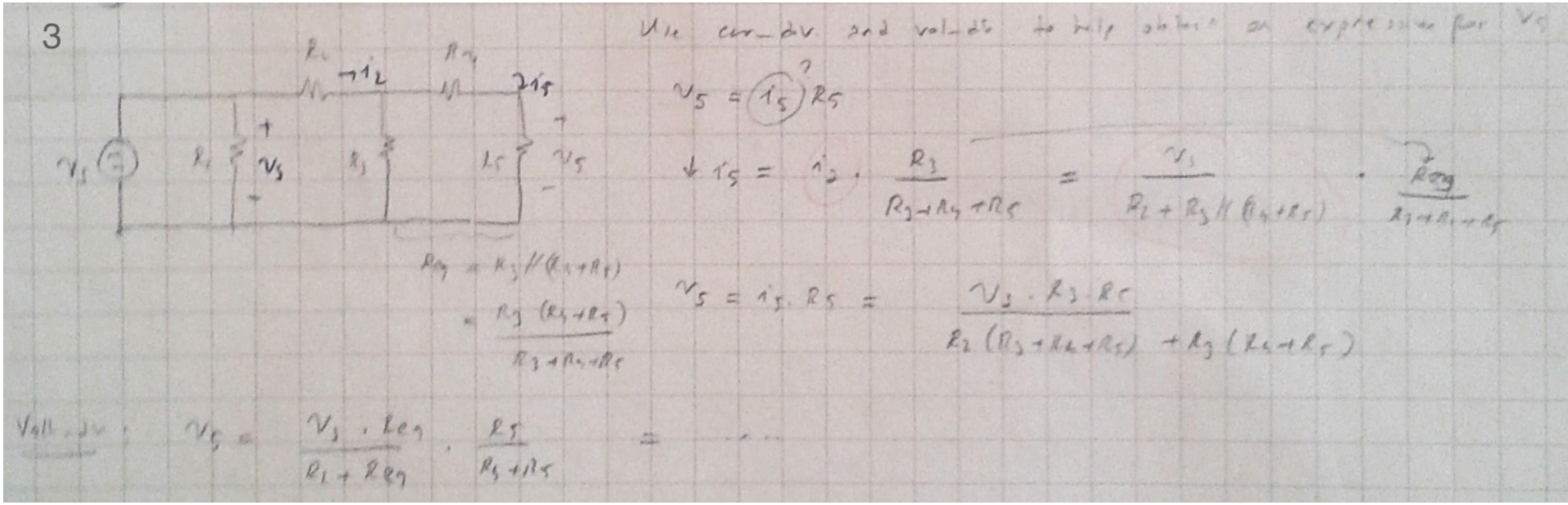
$i_{in} = 1 - 6 - 4 - 4v_x$

$= -49A$.

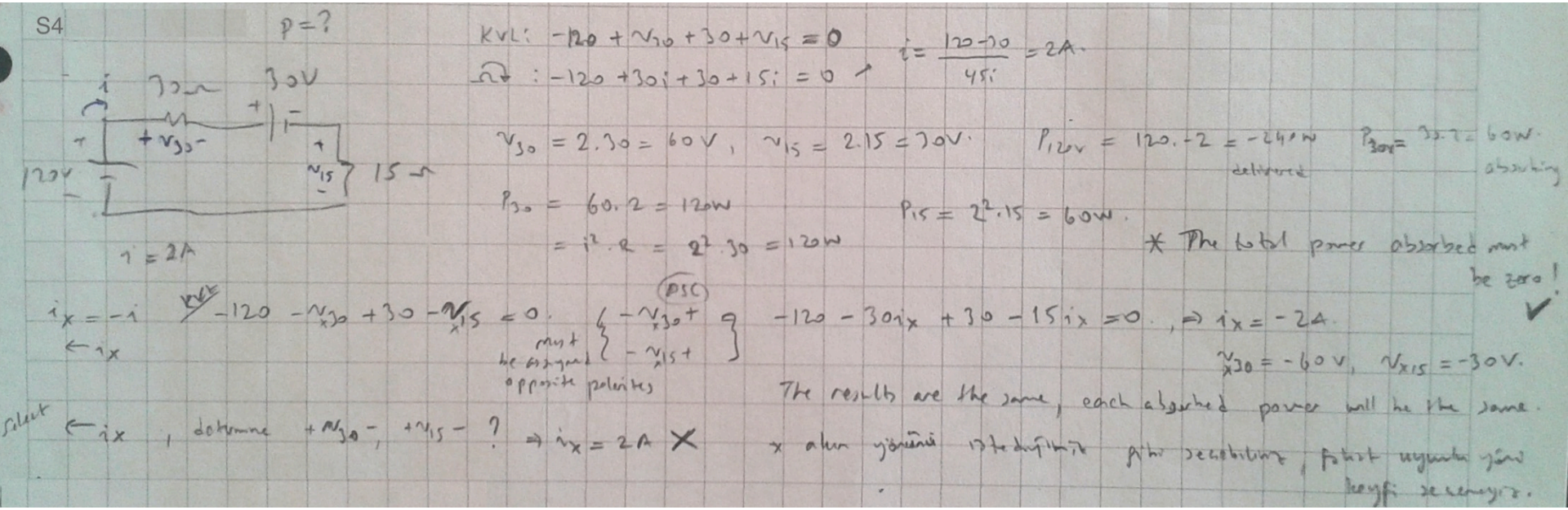
c) $i_{4\Omega} = \frac{10}{4} = 2,5A$ $I_s = -2,5 - 4 - 40 = -46,5A$.

d) $P_{4v_x} = 8 \times 40 = 320W$ → absorbed

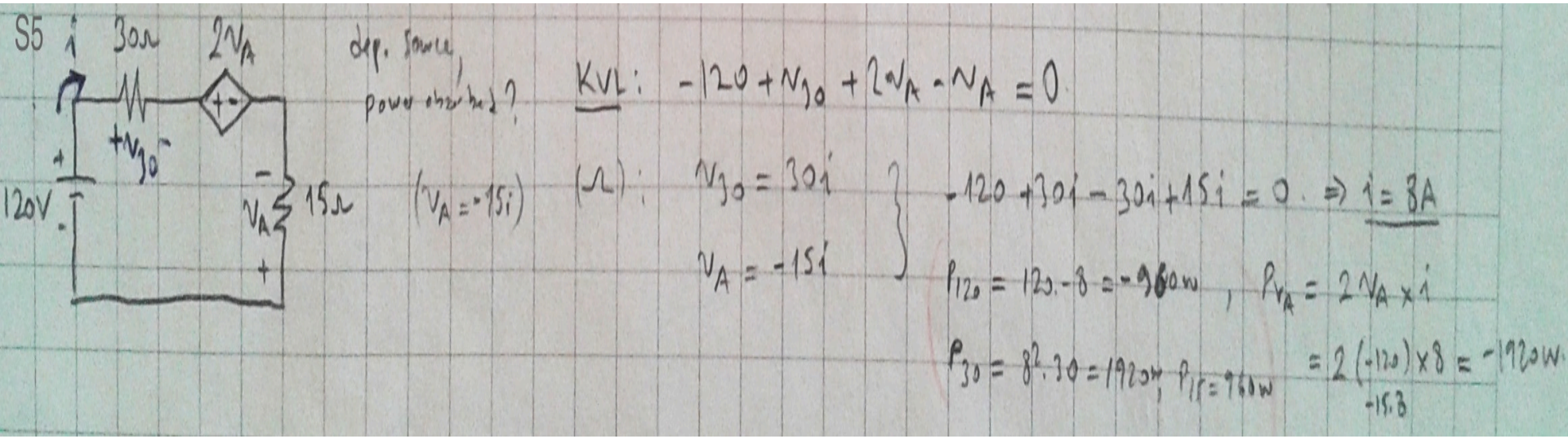
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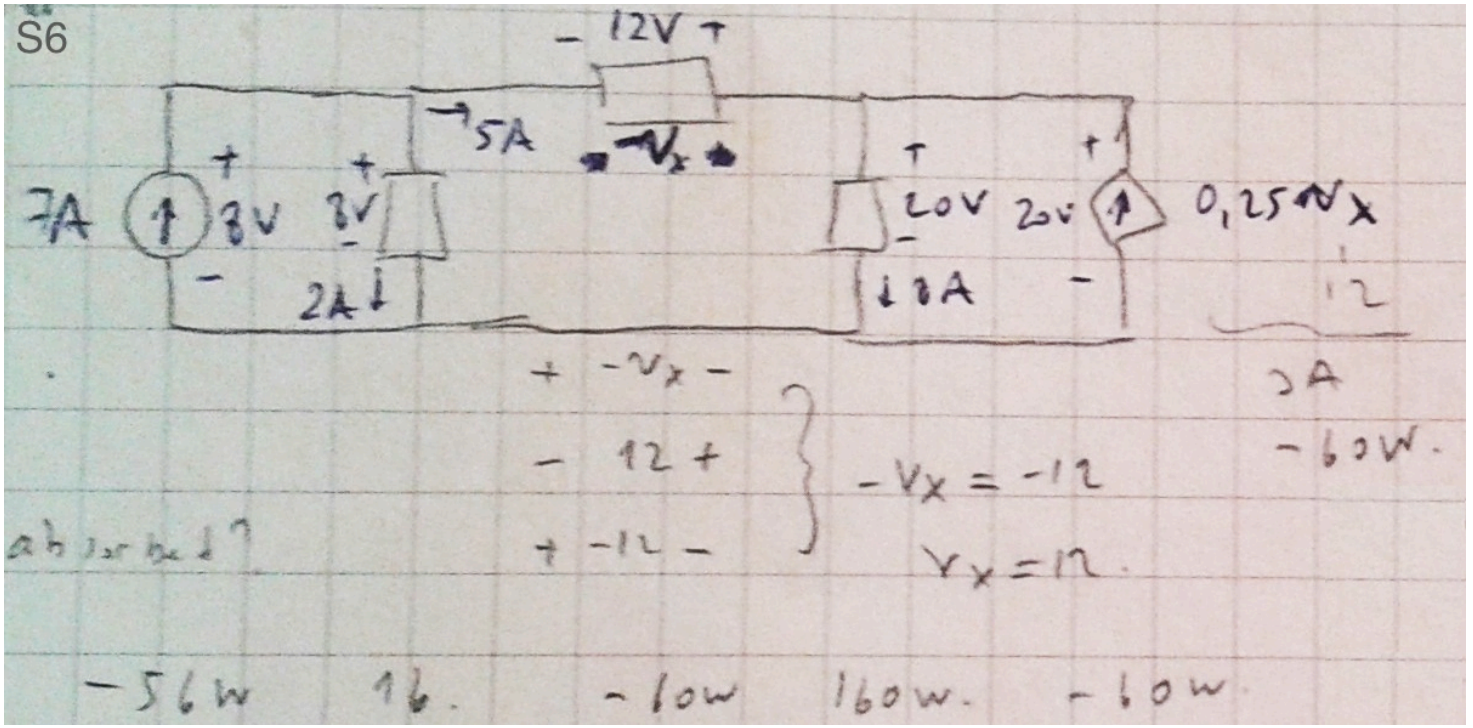
ÖRNEK SORU&ÇÖZÜM



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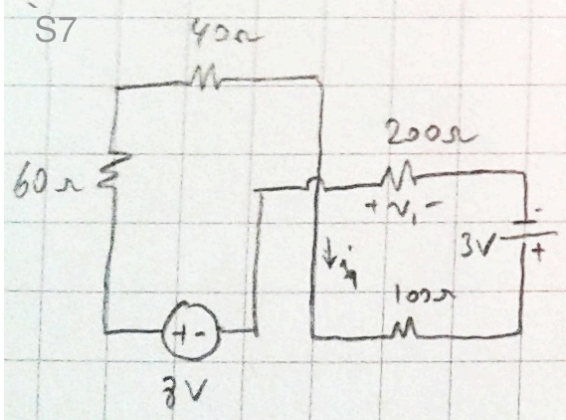


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S7



$i_1, v_1 = ?$

the power supplied by the 3V battery ?

KVL:

$100i_1 + 3 + \underbrace{200i_1}_{-v_1} - 8 + 60i_1 + 40i_1 = 0.$

$i_1 = \frac{8-3}{400} = 12{,}5\,mA.$

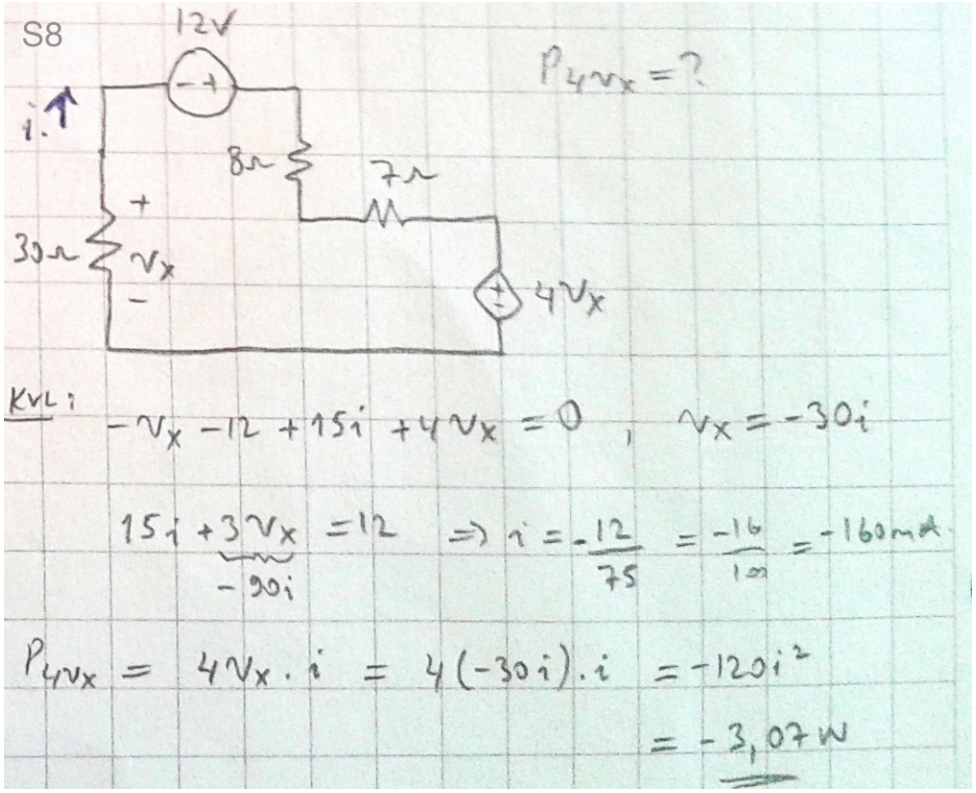
$v_1 = -200i_1 = +200 \times 12{,}5m$

$200 \times (-i_1)$ $= -2{,}5V$

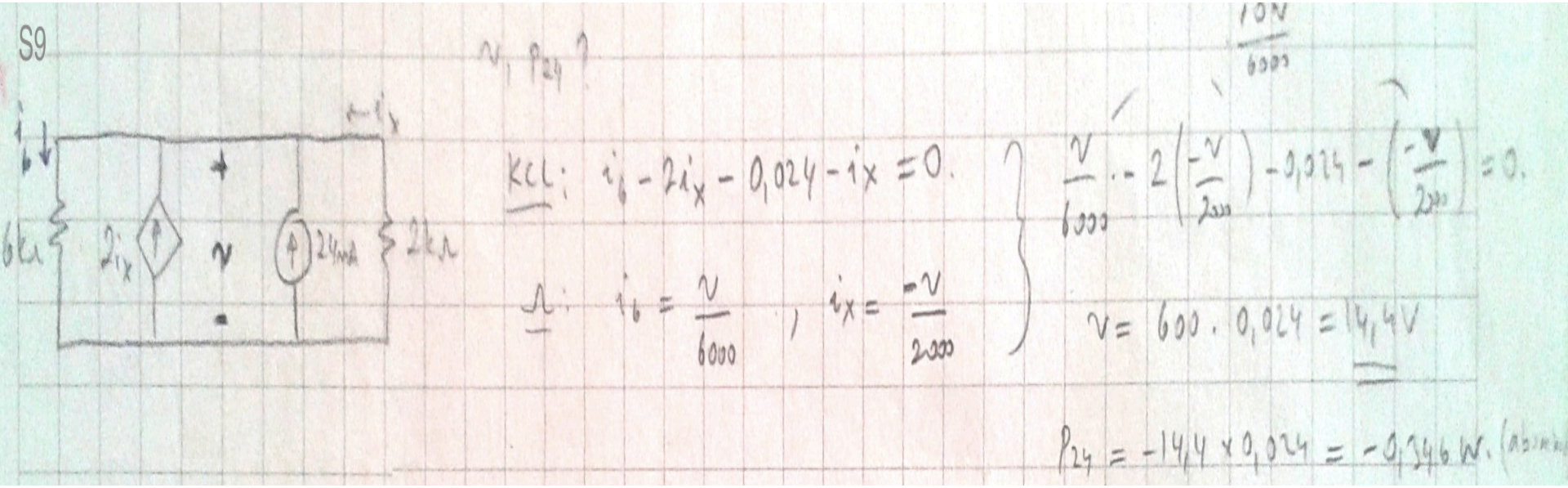
$P_{3V} = 3 \times i_1 = 3 \times 12{,}5m = 37{,}5\,mW.$

supply $\rightarrow -37{,}5\,mW.$

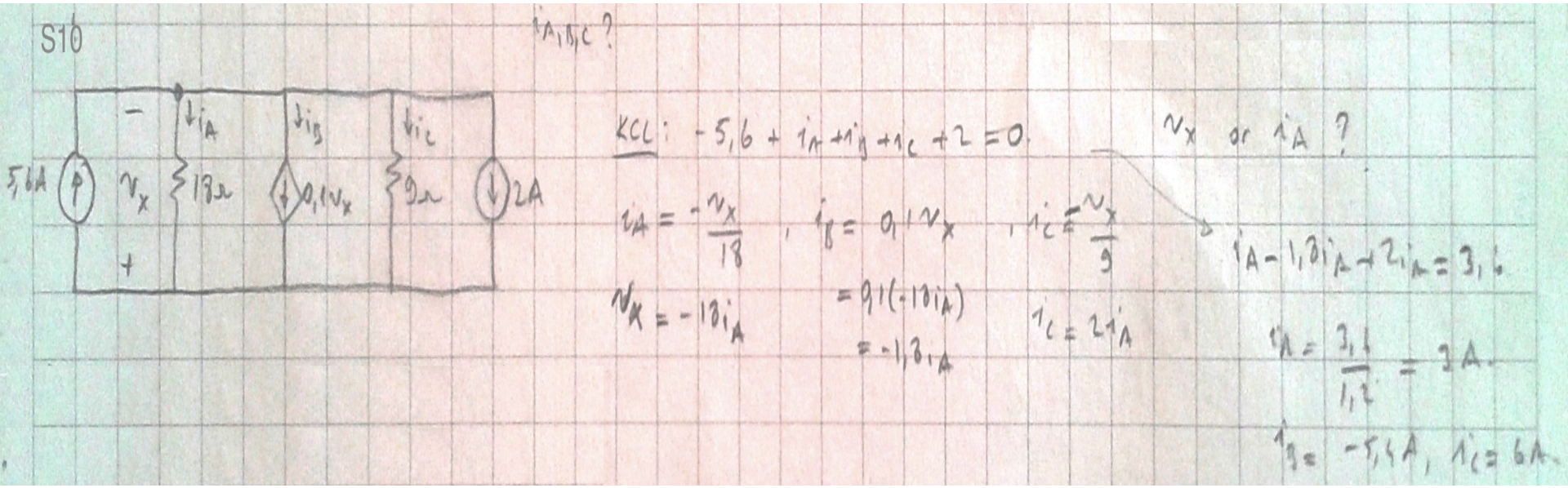
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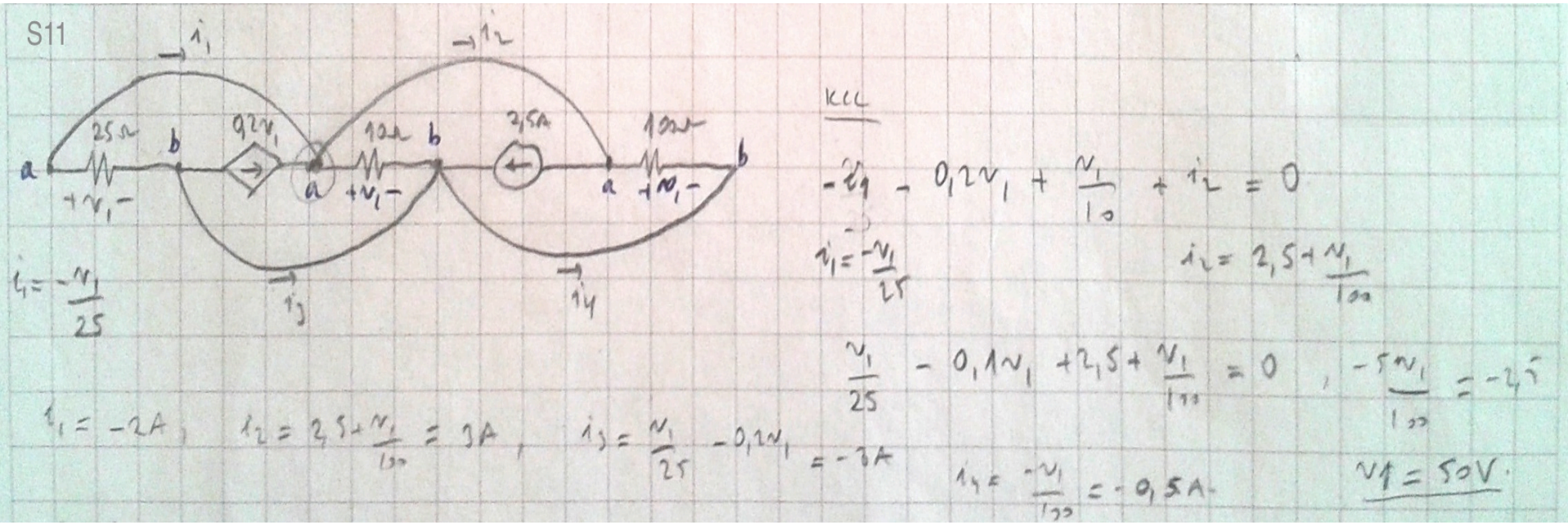
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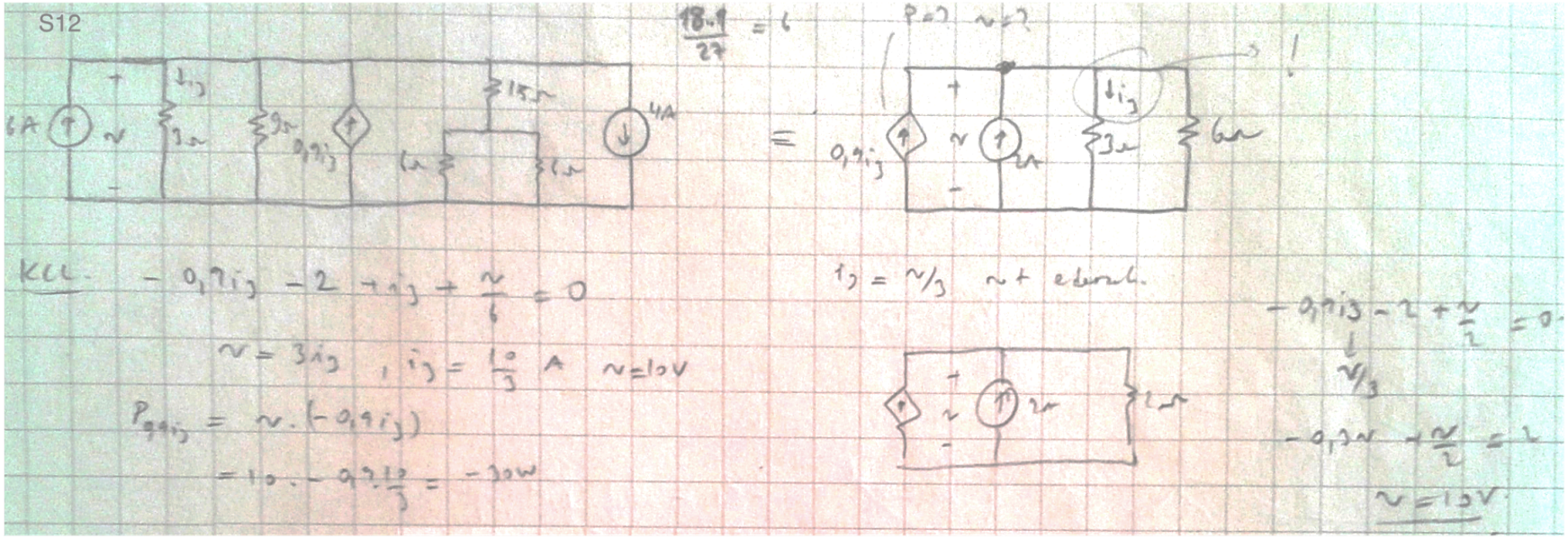
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ÖRNEK SORU&ÇÖZÜM

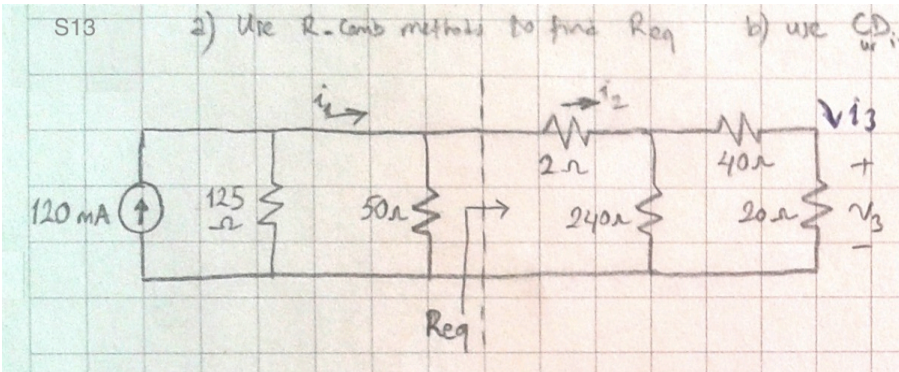


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S13 a) Use R-Comb methods to find R_{eq} b) use CD or iv to find i_1 ? 100 mA, i_2 ? 50 mA, v_3 0,8 V



a) $R_{eq} = [(40+20) \,//\, 240] + 2 = \frac{60 \cdot 240}{300} + 2 = 50\Omega$

b) $i_1 = \frac{125}{125 + 50//50} \cdot 120m = \frac{125}{125+25} 120m = 100mA$ $\quad i_2 = \frac{50}{50+50} \cdot i_1 = 50mA$

(I) CurDiv $\quad i_3 = \frac{240}{300} \cdot i_2 = \frac{4}{5} \cdot 50m = 40mA \Rightarrow v_3 = 20 \cdot 40m = 0,8V$

(II VoltDiv $\quad v_2 = 48\Omega \cdot i_2 = 48 \cdot 50m = 2,4V \Rightarrow v_3 = \frac{20}{40+20} \cdot v_2 = 0,8V$)